Beyond Preference Following: Learning User Decision Preferences for Personal Agents through Interaction

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Abstract

Personalization for large language model (LLM) systems is rapidly moving from response-level style adaptation toward agentic assistance that must make recommendations, invoke tools, and execute actions on behalf of specific users. Recent work has explored personalization via static profiles, long-context memory, user-specific fine-tuning, personalized tool use, and interactive preference elicitation. However, existing methods are evaluated with incompatible objectives: some measure whether a model remembers profile attributes, others test preference following in long context, while agent benchmarks often conflate task completion with personalization quality. We argue that these methods can be unified under a more fundamental objective: *learning a user's decision preference distribution*. Under this view, a personal agent should not merely produce responses that look personalized; it should infer how a particular user makes trade-offs and progressively improve that estimate through interaction.

This paper formulates personal-agent personalization as interactive learning of user decision preferences. We introduce a benchmark and evaluation framework that treats user simulation as a controllable approximation to a family of decision preference distributions, with train/test splits defined over users and preference types rather than over prompts alone. We decompose simulation into a latent decision model and a natural-language verbalizer, define metrics for decision-fit, adaptation efficiency, generalization, and sim-to-real transfer, and propose a human-validation protocol to test whether improvements observed with simulated users transfer to real users. The resulting framework shifts the evaluation target from conversation quality alone to the central capability of personal agents: learning how a user decides.

1 Introduction

Large language models are increasingly deployed as *personal agents*: assistants that do more than answer questions, and instead plan, recommend, invoke tools, and act under user-specific constraints. In this setting, the core challenge is no longer generic helpfulness. A personal agent must reason about a particular user's latent preferences, habits, and trade-offs, then convert that understanding into tailored decisions and actions.

A growing literature has begun to tackle this problem from several directions. Benchmarking work such as *PersonalLLM* focuses on adapting LLM outputs to individual preferences rather than population-level averages [12]. *PrefEval* evaluates whether a model can infer, memorize, and adhere to user preferences in long-context conversations [8]. *PERSONAMEM* and follow-up work emphasize dynamic user profiling and evolving personas across multi-session interactions [1, 9]. At the agent level, *PersonaAgent* studies test-time personalization through memory–action integration [10], while *PEToolLLM* and *PTBench* extend personalization to tool use and parameter selection [6, 3]. Finally, interactive alignment work such as *ALOE* highlights the value of multi-turn interaction

for uncovering implicit user preferences [5], and *UserBench* evaluates agents in preference-driven, underspecified multi-turn interactions [4].

Despite this progress, the field still lacks a unified and objective answer to a basic question: *what exactly should a personal agent be good at?* Existing work often measures one proxy at a time—response-level preference following, memory recall, tool-selection accuracy, or final task success. These are all important, but they do not directly test whether an agent has learned *how a user makes decisions*. This distinction matters. A model can produce personalized language without having learned the user's trade-offs; it can remember profile facts without knowing how those facts should change choices; and it can finish tasks in a simulator without robustly adapting to a real user's preferences.

This problem is compounded by the increasing use of LLM-simulated users in interactive evaluation. Benchmarks such as τ-bench and ToolSandbox scale evaluation by placing an agent in a loop with a simulated user and a tool environment [7, 2]. UserBench similarly relies on user simulation to study preference-driven interaction [4]. However, recent evidence shows that LLM user simulators are often overly cooperative, too stylistically uniform, and weak proxies for human judgments; their downstream evaluations can substantially overestimate real-world performance [11]. Therefore, a stronger evaluation target is needed: not merely whether a method performs well in one simulated dialogue protocol, but whether it can interactively fit a class of user decision preference distributions and generalize across users, preference types, and eventually real humans.

In this paper, we propose to center the problem around **learning user decision preferences for personal agents**. We treat static profile prompting, memory-based personalization, user-specific fine-tuning, personalized tool use, and interactive preference learning as different mechanisms for estimating a conditional user-specific decision distribution. Within this framing, interaction is not the problem itself; it is the mechanism that allows the agent to progressively reduce uncertainty over that distribution. This yields a more coherent research agenda and a more rigorous benchmark design.

Contributions. We make four contributions:

1. We formulate personalization for personal agents as *interactive estimation of a user decision preference distribution*, shifting the focus from response style to decision making.

2. We introduce a benchmark design in which user simulation is decomposed into a *decision generator* and a *language verbalizer*, reducing confounds between surface style and latent decision preference.

3. We define a unified evaluation protocol based on *decision-fit*, *adaptation efficiency*, *cross-preference generalization*, and *sim-to-real transfer*.

4. We propose a human-validation methodology that tests whether simulator rankings and adaptation gains transfer to real users, directly addressing sim-to-real concerns in interactive user simulation [11].

2 Related Work

2.1 Personalization in LLMs

A major recent trend in personalization moves beyond universal alignment and toward user-specific adaptation. *PersonalLLM* introduces a benchmark for tailoring LLM responses to individuals,

emphasizing that heterogeneous user preferences cannot be captured by a single global alignment target [12]. *PrefEval* evaluates whether models can infer, memorize, and adhere to user preferences in long-context conversations, showing that preference following is a multi-stage competency rather than a single task [8]. *PERSONAMEM* expands the focus to dynamic user profiling, measuring whether a model can track evolving personas and apply them to later responses [1]. In parallel, *RLPA* introduces dynamic profile modeling through simulated interaction, arguing that static profile conditioning is insufficient for long-term personalized alignment [9]. These works establish the importance of individualized modeling, but they primarily evaluate text generation or profile adherence rather than action-level decision competence.

2.2 Personalized agents and tool use

As LLM systems become agentic, personalization must affect not just language but also actions. *PersonaAgent* proposes a memory–action framework for sustaining personalized multi-turn interactions at test time, highlighting the need to integrate memory retrieval with downstream decision making [10]. *PEToolLLM* formulates personalized tool learning, where tool invocation depends on both explicit instructions and implicit user preferences reflected in interaction history [6]. *PTBench* extends this line by separating personalized tool selection from personalized parameter inference, making the benchmark target closer to action-conditioned personalization [3]. These efforts are highly relevant to personal agents, but their evaluation remains largely domain-specific and does not yet provide a general framework for comparing static and interactive personalization mechanisms.

2.3 Interactive preference alignment and user-centric agent benchmarks

Interaction can reveal implicit preferences that are unavailable from a static profile. *ALOE* directly studies aligning LLMs with individual preferences through multi-turn interaction, providing evidence that models can improve by explicitly learning to ask and infer personalized constraints over time [5]. *UserBench* evaluates multi-turn, preference-driven user interactions with underspecified goals and incrementally revealed constraints, further underscoring the importance of active clarification in user-centric agents [4]. More generally, interactive benchmarks such as τ-bench and ToolSandbox expose agents to users, tools, and environment state in the loop [7, 2]. Yet these environments were not designed as a unified test of *decision preference learning*; the evaluation target is often task completion or benchmark-specific reward.

2.4 User simulation and the sim-to-real gap

Scaling interactive evaluation usually requires simulated users, but recent work cautions against treating simulators as faithful stand-ins for humans. *Mind the Sim2Real Gap in User Simulation for Agentic Tasks* shows that LLM-based user simulators are systematically more cooperative, less frustrated, and more positive in evaluation than real humans, creating an "easy mode" that inflates agent performance [11]. This finding suggests that user simulation should not be treated as a final substitute for human evaluation. Instead, a simulator should be used as a controllable environment for probing what an agent has learned, with explicit human validation reserved for external validity. Our work adopts this position and makes sim-to-real transfer a first-class evaluation criterion.

3 Problem Formulation

We model a personal-agent interaction as a sequential decision process involving a user $u \in \mathcal{U}$, a task context $x \in \mathcal{X}$, a history $h_t \in \mathcal{H}$ at turn t, and a decision variable $y_t \in \mathcal{Y}(x, h_t)$ that represents the user's preferred action, recommendation, ranking, or choice under the current state.

Definition 1 (User decision preference distribution)**.** *For a user u, let $z_u \in \mathcal{P}$ denote latent decision-preference variables, such as price sensitivity, risk tolerance, convenience preference, privacy sensitivity, brand loyalty, or fairness constraints. The user's decision behavior is a conditional distribution*

$$p_u(y_t \mid x, h_t) \;=\; p(y_t \mid x, h_t, z_u), \tag{1}$$

which specifies how the user decides under a task context and an evolving interaction history.

This formulation intentionally treats preference as a *distribution*, not a single fixed label. First, users may behave stochastically or inconsistently under ambiguity. Second, the same user may respond differently as new information is revealed. Third, the agent's job is often to estimate uncertainty over future user decisions, not merely to predict one canonical response.

An assistant agent interacts with the user by selecting conversational actions $a_t \in \mathcal{A}$, such as asking a clarification question, proposing a recommendation, invoking a tool, or committing to an action. The user replies with an observation o_t that may include natural language and, in some tasks, an explicit or implicit decision signal. The history updates as

$$h_{t+1} = h_t \oplus (a_t, o_t). \tag{2}$$

The agent maintains a belief state $q_t(z_u)$ over the user's latent preferences and updates it online:

$$q_{t+1}(z_u) \propto q_t(z_u)\, p(o_t \mid a_t, x, h_t, z_u). \tag{3}$$

Although real systems may instantiate this update implicitly through an LLM, memory module, retrieval pipeline, or policy network, the core objective is the same: progressively reduce uncertainty over the user's decision preferences.

We therefore define personalization quality not as stylistic similarity alone but as **decision-fit under interaction cost**. Let $\hat{p}_\theta(y \mid x, h_t)$ denote the agent's predictive distribution over the user's decision after interacting up to turn t. A generic training or evaluation objective is:

$$\mathcal{L}(\theta) = \mathbb{E}_{u,x}\left[\ell(\hat{p}_\theta(\cdot \mid x, h_T), p_u(\cdot \mid x, h_T)) + \lambda\, C(h_T)\right], \tag{4}$$

where ℓ is a proper scoring rule or ranking loss, $C(h_T)$ measures interaction cost (e.g., number of turns, API calls, or user burden), and λ trades off fit against efficiency.

This perspective unifies diverse personalization mechanisms:

- **Static profile prompting** initializes $q_0(z_u)$ from a fixed profile.
- **Memory-augmented methods** store partial sufficient statistics of past interactions to update q_t.
- **User-specific fine-tuning** amortizes preference estimation into model parameters.
- **Interactive preference learning** actively chooses a_t to reveal high-information aspects of z_u.

All of them can be compared by how well they fit the user's decision distribution and how efficiently they adapt.

4 Benchmark Design

We introduce a benchmark framework, **Decision Preference Learning for Personal Agents** (DPL-PA), designed to evaluate whether an agent can learn user decision preferences through interaction. The framework is intentionally modular: it can be instantiated in recommendation, travel planning, shopping, scheduling, education, or tool-use domains, as long as each task requires the user to make meaningful trade-offs.

4.1 Benchmark instance structure

Each benchmark instance contains five components:

1. **Task context** x: a structured or semi-structured scenario, such as a set of candidate products, flights, itineraries, calendars, or tools.
2. **Candidate decision space** $\mathcal{Y}(x)$: the set of options, rankings, actions, or tool invocations that the user might choose.
3. **User representation**: a profile p_u^{text} and latent preference variables z_u.
4. **Interaction channel**: a multi-turn natural-language protocol in which the agent may ask questions, make proposals, and receive user responses.
5. **Ground-truth decision process**: one or more decision labels or distributions associated with the user under the task context.

The benchmark is not tied to one response format. For some tasks, y is a discrete option; for others it may be a ranking, a binary accept/reject label, or a structured action such as a tool call with parameters.

4.2 Decomposing user simulation

A central design choice is to *separate latent decision generation from linguistic realization*. Existing simulated-user benchmarks often rely on a single LLM prompt to both decide and speak like a user. This entangles true decision preference with language style, cooperation level, verbosity, and other superficial artifacts. Following recent sim-to-real findings [11], we instead decompose simulation into two modules:

$$y_t \sim f_{\text{dec}}(z_u, x, h_t), \tag{5}$$

$$o_t = f_{\text{lang}}(z_u, x, h_t, y_t, \epsilon_t), \tag{6}$$

where f_{dec} is a **decision generator** and f_{lang} is a **language verbalizer**. The former determines what the user chooses; the latter determines how that choice is expressed.

This decomposition has three advantages. First, it permits direct validation of decision diversity independent of style. Second, it allows controlled perturbations of language without changing ground-truth preferences. Third, it provides access to latent decision labels at every step, enabling stronger evaluation of the assistant's preference model.

4.3 Constructing diverse preference types

We define preference types as structured regions in latent preference space $\mathcal{P}$. In a travel domain, for instance, z_u may include dimensions such as cost sensitivity, delay tolerance, layover aversion, loyalty-program strength, seat comfort preference, and baggage flexibility. In a shopping domain, z_u may include price sensitivity, brand preference, sustainability preference, and urgency. In a scheduling domain, z_u may include time flexibility, commute aversion, fairness constraints, and privacy preferences.

Rather than assuming that one user equals one rigid type, we represent each user as a *conditional preference distribution* over these dimensions. Preference types can then be defined either by clustering historical user decisions or by controlled generation from a latent-template library. This supports multiple train/test regimes:

- **User-disjoint split**: train and test users are disjoint.
- **Preference-cluster-disjoint split**: train and test do not share preference clusters.
- **Context-shift split**: users are held out under new task contexts to test whether the agent learned preferences rather than template-specific actions.

4.4 Validation of simulator diversity

The benchmark requires explicit simulator validation before it can be used to compare agents. We propose four validation tests:

1. **Within-profile stability**: the same profile should lead to stable decisions on semantically equivalent tasks.
2. **Across-profile separability**: distinct profiles should induce significantly different decision distributions under the same context.
3. **Counterfactual sensitivity**: modifying one preference dimension should mainly affect decisions related to that dimension.
4. **Paraphrase invariance**: changing the linguistic surface form of a task should not substantially alter the underlying decision distribution.

These checks are necessary to ensure that the simulator is measuring preference learning rather than prompt sensitivity or stylistic mimicry.

5 Evaluation Protocol

We evaluate a personal agent along four axes: decision-fit, adaptation efficiency, generalization, and sim-to-real transfer.

5.1 Decision-fit

The primary question is whether the agent has learned how the user decides.

Pointwise decision consistency. For categorical decisions, we measure exact-match accuracy and Cohen's κ between the agent's predicted decision and the user's realized decision. When multiple valid decisions exist, we additionally report top-k accuracy.

Distributional fit. Since preferences are modeled as distributions, we report proper scoring rules such as negative log-likelihood (NLL) and Brier score. Let $\hat{p}_\theta(y \mid x, h_t)$ denote the agent's predictive distribution and let $y_t^\star$ denote the realized user decision. Then

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_\theta(y_i^\star \mid x_i, h_{i,T}). \tag{7}$$

We also compute calibration error over decisions to test whether the agent's confidence matches the user's observed choices.

Ranking fit. For recommendation or option-ranking tasks, we compute Spearman correlation, Kendall's τ, and NDCG between the agent's ranking and the user's latent or observed ranking.

5.2 Adaptation efficiency

A personal agent should not only fit the user; it should do so efficiently.

Turn-to-threshold. Given a target decision-fit threshold α, we define

$$T_\alpha = \min\{t : \mathrm{Fit}(h_t) \geq \alpha\}. \tag{8}$$

This measures how many interaction turns are required before the agent reaches acceptable adaptation quality.

Interaction-quality curves. We report area under the consistency–turn curve (AUC-CT) and area under the NLL–turn curve. These summarize how quickly an agent improves over the course of interaction.

User-cost-aware efficiency. Not all turns impose equal burden. We therefore define a weighted cost function

$$C(h_T) = \sum_{t=1}^{T} c(a_t, o_t), \tag{9}$$

where c can depend on the number of clarification questions, intrusive questions, tool calls, or correction requests. This captures the practical reality that a personal agent should minimize user effort.

5.3 Generalization

We explicitly separate three forms of generalization:

- **Unseen-user generalization**: adapting to new users from seen preference regions.
- **Unseen-preference generalization**: adapting to users whose preferences fall in held-out regions or clusters.
- **Cross-context generalization**: transferring the learned preference model to new decision contexts for the same user.

This decomposition avoids overclaiming generalization from user memorization alone.

5.4 Sim-to-real transfer

Since simulation is only a proxy, we treat transfer to real users as a first-class metric.

Human gap. Let M denote a method and $\mathcal{S}$ and $\mathcal{H}$ denote the simulated and human settings. For any score s, define the human gap

$$\Delta_{\mathrm{H}}(M) = s(M; \mathcal{S}) - s(M; \mathcal{H}). \tag{10}$$

A small gap indicates that simulator gains transfer better to humans.

Rank transfer. A useful simulator should preserve method ordering. We therefore compute Spearman correlation between method rankings on simulator-based evaluation and human evaluation.

Behavioral fidelity diagnostics. We separately assess whether the simulator reproduces human decision diversity, uncertainty, correction behavior, and non-cooperation, building on recent sim-to-real findings for user simulation [11].

6 Agents and Baselines

The framework is designed to compare personalization methods that operate under different information and learning regimes. We recommend at least the following baseline families.

Static profile prompting. The agent receives a profile summary and answers without persistent belief updates. This baseline tests whether fixed profile information is sufficient.

Retrieval and memory augmentation. The agent retrieves relevant user history or memory snippets at test time. This represents a common strategy in memory-augmented personal agents.

User-specific fine-tuning or lightweight adaptation. The agent is adapted to user-specific data through parameter-efficient tuning, reward modeling, or user-conditioned heads. This captures amortized personalization.

Interactive belief tracking. The agent explicitly maintains a latent user model and chooses clarification questions to maximize information gain. This family operationalizes interactive preference learning most directly.

Action-level personalization baselines. For tool or decision-action tasks, we include methods inspired by PEToolLLM and related personalized tool invocation systems, which infer latent user constraints from history when selecting tools or parameters [6, 3].

7 Experimental Design

This paper proposes a benchmark and evaluation framework rather than reporting completed empirical results. To support a rigorous future submission, we outline the experimental protocol required for full evaluation.

7.1 Research questions

The benchmark is designed to answer the following questions:

- **RQ1:** Which personalization mechanisms best fit user decision preferences under equal interaction budgets?
- **RQ2:** Does interaction improve personalization beyond static profile conditioning and memory retrieval?
- **RQ3:** Do gains on simulated users transfer to real users?
- **RQ4:** Which kinds of preferences are easiest or hardest to learn interactively?

7.2 Task domains

A strong benchmark should include multiple decision-centric domains in which personalization changes actions rather than only wording. Suitable domains include:

1. **Travel planning**: selecting itineraries under cost, time, comfort, and risk trade-offs.
2. **Shopping**: choosing products under price, brand, sustainability, quality, and urgency trade-offs.
3. **Scheduling**: selecting meeting times or plans under convenience, fairness, privacy, and commute constraints.
4. **Tool use**: choosing among functionally similar tools or inferring missing parameters from user preference history.

Each domain should expose both explicit constraints and latent user-specific trade-offs so that interaction is useful but not trivially mandatory.

7.3 Simulator construction protocol

We recommend a two-stage simulator construction pipeline:

1. **Latent preference design**: define domain-relevant preference dimensions and either sample or estimate z_u from user histories.
2. **Language realization**: condition an LLM verbalizer on (z_u, x, h_t, y_t) to produce realistic user utterances while preserving the ground-truth decision process.

This design permits intervention on language style independently of latent decision preference, which is critical for disentangling what the agent has learned.

7.4 Human validation protocol

To test external validity, we propose a human study in which participants interact with the same agent methods on matched tasks. Each participant is assigned either a real profile created from pre-study elicitation or a minimal profile plus in-session decisions. We then compare:

1. simulator-based and human-based method rankings;
2. simulator-based and human-based adaptation curves; and

3. simulator-based and human-based failure modes.

This allows the benchmark to support the strongest plausible claim: not that a simulator replaces humans, but that simulator-based evaluation can be *validated* as a proxy for decision preference learning.

8 Why This Benchmark Is Needed

We conclude this methodology section by clarifying the gap filled by the proposed benchmark.

Existing personalization benchmarks mostly test response quality or memory. PersonalLLM and PrefEval are highly valuable for user-specific response personalization, but they are not explicitly designed around decision distributions or action-level trade-offs [12, 8]. PERSONAMEM and RLPA emphasize evolving user profiles, but the central evaluation target remains profile tracking and personalized response generation [1, 9]. Our benchmark instead centers the object being learned: the user's decision preference distribution.

Existing agent benchmarks mostly test task completion. Interactive environments such as τ-bench, ToolSandbox, and UserBench capture multi-turn tool or preference-driven interaction, but their success metrics are still dominated by task completion, benchmark-specific reward, or underspecification resolution [7, 2, 4]. These are necessary but not sufficient for evaluating personal agents. An agent may succeed at a task while poorly modeling the user's preferences, or fail to ask the right questions while still producing plausible outputs.

A unified framing enables fairer comparison across method classes. Once personalization is defined as estimating a user decision distribution under interaction cost, profile prompting, retrieval, memory, fine-tuning, interactive alignment, and personalized tool use can be compared within the same framework. This is useful both for benchmarking and for scientific understanding: it reveals when interaction helps, when static history is enough, and which kinds of personalization remain hard.

9 Limitations and Ethical Considerations

The proposed framework has several limitations.

First, even with latent decision generators, simulated users remain approximations to real people. Human validation is therefore mandatory, not optional. Second, designing preference dimensions requires normative choices; poorly chosen dimensions may erase meaningful diversity or encode unwanted stereotypes. Third, collecting user histories and personalization data raises privacy concerns. Any benchmark instantiation should adopt strict data minimization, informed consent, and user-level deletion controls. Fourth, personal agents that optimize too aggressively for inferred preferences may over-personalize, manipulate, or prematurely commit to inferred values. Evaluation should therefore include uncertainty calibration and conservative behavior under low confidence.

From an ethical perspective, a strong personal agent should learn preferences *with the user*, not merely *about the user*. This motivates interactive designs that expose uncertainty, ask clarifying questions when needed, and allow users to override stored inferences.

10 Conclusion

We presented a new framing for personalization in personal agents: *learning user decision preferences through interaction*. This framing unifies recent work on static profiles, long-context memory, user-specific adaptation, personalized tool use, and interactive alignment under a common objective: estimating a user-specific decision preference distribution. Based on this perspective, we proposed a benchmark and evaluation framework that (i) decomposes user simulation into latent decision generation and language realization, (ii) defines train/test splits over users and preference types, (iii) evaluates decision-fit, adaptation efficiency, generalization, and sim-to-real transfer, and (iv) requires explicit human validation. We believe this shift—from surface personalization to decision preference learning—provides a more principled foundation for personal-agent research and a stronger route toward trustworthy personalization in real-world interactive systems.

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